

2. Project Requirements Analysis

Project Name

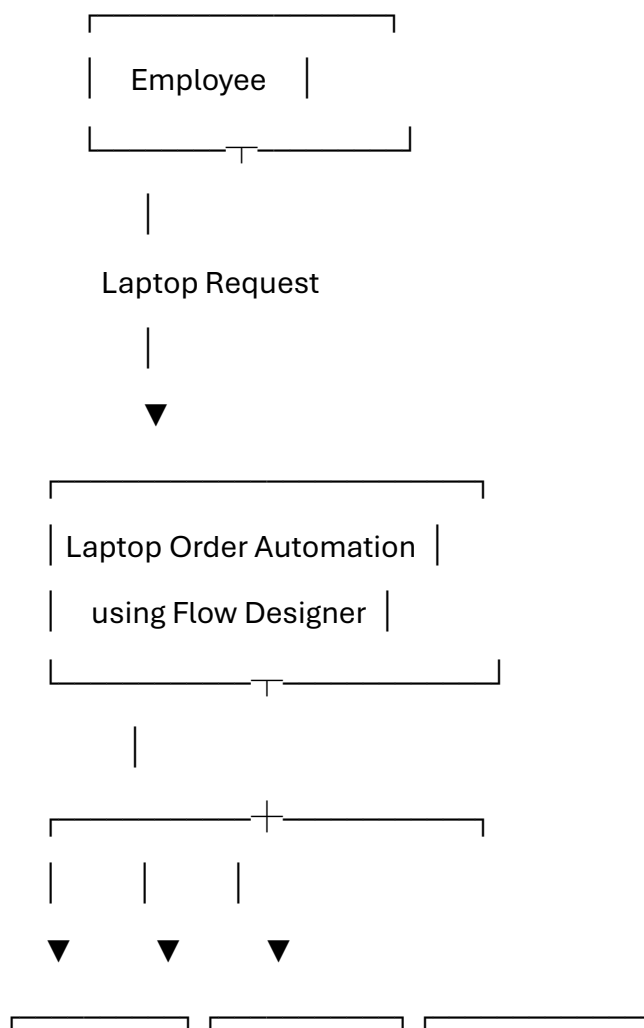
Automating Standard Laptop Orders with Flow Designer

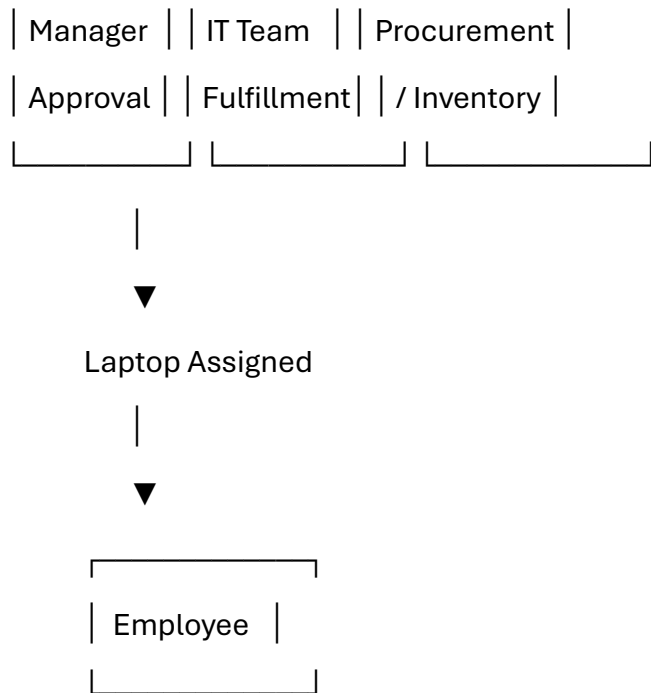
Team Members

- **Team Leader:** SRIPRIYA L
 - **Team Member:** HARINI K
 - **Team Member:** SATHIYA PRIYA M
 - **Team Member:** SIVARANJANI S
 - **Team Member:** SAKTHISRI P
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2.1 Data Flow Diagram (DFD)

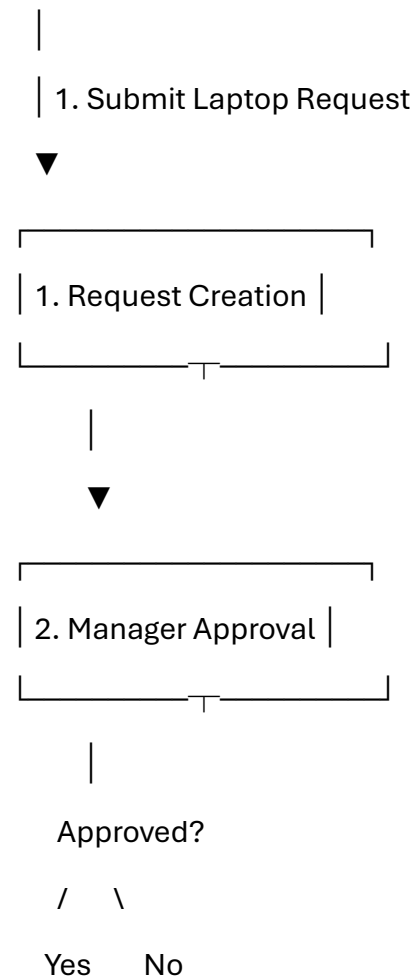
Level 0 – Context Diagram





Level 1 – Detailed DFD

Employee

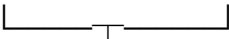




Notify Employee

3. Check

Availability



Available?



Yes No



Create IT Notify /

Fulfillment Procurement

Task



Laptop Assignment



Update Asset Record



Notify Employee



Close Request

Main Data Stores

- **D1 – User/Employee Database**
- **D2 – Laptop Inventory/Asset Database**
- **D3 – Laptop Request Records**
- **D4 – Approval Records**
- **D5 – Fulfillment Task Records**

2.2 User Stories

ID	User Story	Acceptance Criteria
US01	As an employee, I want to submit a standard laptop request so that I can receive a laptop for my work.	Request can be submitted with required details.
US02	As a manager, I want to approve or reject laptop requests so that I can control equipment allocation.	Manager can approve/reject the request.
US03	As the system, I want to automatically route requests to the appropriate manager so that manual routing is avoided.	Request reaches the correct approver automatically.
US04	As the IT team, I want an automatic fulfillment task so that I can process approved requests.	IT task is created after approval.
US05	As the system, I want to check laptop availability so that only available devices are assigned.	Inventory is checked automatically.
US06	As an employee, I want to receive notifications about my request so that I know its current status.	Notifications are sent for major status changes.
US07	As the IT team, I want the asset record to be updated after assignment so that inventory remains accurate.	Assigned laptop is linked to the employee.
US08	As an employee, I want to track my request so that I do not need to contact IT repeatedly.	Current request status is visible.

2.3 Solution Requirements

Functional Requirements

1. The system must provide a laptop request form.
2. The requester must enter required employee and laptop information.
3. The system must automatically identify the appropriate approver.
4. The request must be routed for manager approval.
5. The system must support **Approve** and **Reject** actions.
6. The system must automatically check laptop availability.
7. An IT fulfillment task must be created after approval.
8. The system must send notifications to the requester.
9. The laptop asset record must be updated after assignment.
10. The request must automatically move to completion after successful fulfillment.
11. Rejected requests must be closed with an appropriate status.
12. Request and fulfillment information must be stored for tracking and reporting.

Non-Functional Requirements

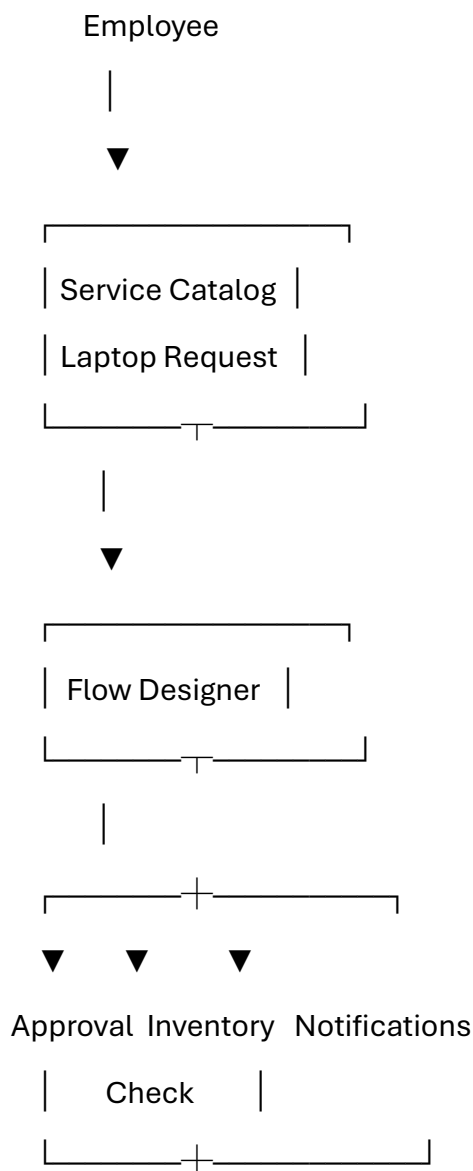
- **Reliability:** The workflow should execute consistently without losing requests.
- **Security:** Only authorized users should access employee and asset information.
- **Performance:** Requests should be processed quickly after submission.
- **Usability:** The request form should be simple and easy to understand.
- **Scalability:** The solution should support increasing numbers of laptop requests.
- **Auditability:** Approvals, assignments, and status changes should be recorded.
- **Maintainability:** The Flow Designer workflow should be easy to modify when business rules change.

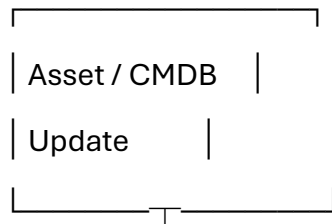
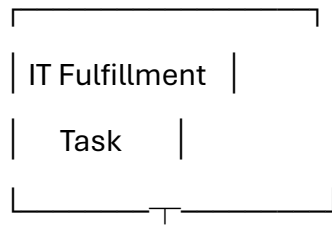
2.4 Technology Stack

Technology	Purpose
ServiceNow	Main platform for implementing the laptop-ordering solution
Flow Designer	Automates the complete laptop request workflow
Service Catalog	Provides the laptop request form/catalog item

Technology	Purpose
ServiceNow Tables	Stores request, user, task, and asset information
CMDB / Asset Management	Maintains laptop and asset information
Notifications	Sends request and approval status updates
Approvals	Handles manager approval and rejection
JavaScript	Can be used for custom logic where required
ServiceNow Reports/Dashboards	Used to monitor requests and workflow performance

Technology Architecture





Laptop Assigned



Request Closed